Hanting Ye

Postdoc · Marie-Curie PhD Fellow
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Research Interests

Physics-Informed Machine Learning, Mobile Computing/Systems, AR/VR/XR, Wireless Communications

I am passionate about leveraging ubiquitous mobile devices as multisensory interfaces to enable mobile AI systems that can perceive, understand, and interact with users and the physical world. I have identified and formalized several types of useful physical information in the optical domain by leveraging pervasive components such as displays and cameras across emerging mobile-device form factors. I am also interested in how different types of physical information interact to generate new insights for applying AI to specific tasks, while ensuring the efficient, responsive, robust, and trustworthy operation of deployed mobile systems in real-world environments. Moving forward, I aim to enable multimodal perception on emerging egocentric mobile platforms, e.g., smart glasses, and leverage physical insights to augment AI and robots and support humans in applications such as personal health management.

Education and Professional Experience

Duke University — *Durham, United States*
Postdoctoral Associate, Intelligent Interactive Internet of Things (I3T) Lab, Department of Electrical and Computer Engineering — *05/2025 – Present*
- Advisor: Prof. Maria Gorlatova

Delft University of Technology (TU Delft) — *Delft, Netherlands*
PhD with *Cum Laude* (**the highest distinction, top 5%**), Embedded Systems (ES) Group, Computer Science Department — *10/2020 – 04/2025*
- Thesis: *Through-Screen Computing*
- Promotors: Prof. Qing Wang, Prof. Marco Antonio Zúñiga Zamalloa

PureLiFi Technology — *Edinburgh, United Kingdom*
Visiting PhD Student — *09/2022 - 01/2023*
- Advisor: Prof. Harald Haas

University of Electronic Science and Technology of China (UESTC) — *Chengdu, China*
M.S. in Electronics and Communication Engineering, National Key Laboratory of Wireless Communication — *09/2017 – 06/2020*
- Advisor: Prof. Xin Kang

Chongqing University (CQU) — *Chongqing, China*
B.S. in Communication Engineering with Honors, Hongshen Honors School — *09/2013 – 06/2017*

Honors & Awards

ACM HotMobile Best Demo Award — *2026*
(Top 5%, the highest distinction) Doctoral Award with Cum Laude of TU Delft — *2025*
(100 total in computer science per year) Heidelberg Laureate Forum Young Researcher — *2024*
A week-long forum connecting selected young researchers with ACM Turing Award and ACM Prize in Computing laureates.
(Top 1%) Excellent Student Leader Award of UESTC — *2019*
(Top 1%) Outstanding Student Awards of UESTC — *2018, 2019*
(Top 1%) First-Class Master's Degree Scholarship — *2018, 2019*
(Top 1%) 1st Prize, 15th China Graduate Mathematical Contest in Modelling — *2018*
(Top 0.5%) Outstanding Graduate Award of Chongqing Province — *2017*
(Top 1%) Outstanding Graduates Awards of CQU — *2017*
(Top 3%) Enrollment Scholarship for Master's Students — *2017*
(Top 1%) Outstanding Student Awards of CQU — *2016*
(Top 5%) Honorable, International Mathematical Contest in Modeling — *2016*
(Top 1%) National Scholarship — *2016*
(Top 1%) 1st Prize, China Undergraduate Mathematical Contest in Modelling — *2015*
(Top 5%) Honorable, Electronic Skills Competition of CQU — *2015*
(Top 3%) National Inspirational Scholarship — *2015*
(Top 5%) Second-Class Bachelor's Degree Scholarship — *2014, 2015*

(Top 1%) 1st Prize, Mobile app design competition of CQU *2014*

Grants & Fellowships

HCC: Foundations of Automated Evaluation and Monitoring of AR Guidance in Pervasive XR and Human-Robot Collaboration (HRC) Applications — **NSF Future CoRe** (Funded) 500 K$ *US · 02/2026*
- PI: Maria Gorlatova; Co-PI: Miroslav Pajic
- Contributions: Contributed to writing one technical thrust and developed visualizations for other thrusts.

Project Aria Gen 2 Glasses for Facilitating Human-Robot Collaboration — **Meta** (Funded) 5 glasses sets *US · 12/2025*
- PI: Maria Gorlatova
- Contributions: Led the conceptualization and writing of the proposal, mentoring a PhD student in the process.

European Marie-Curie PhD Fellowship — **European Union** · ≈128 K€ · *2022*

Abbe Grant — **Carl-Zeiss-Stiftung** · *2024*

Travel Grants Heidelberg Laureate Forum 2024, EWSN 2024, ACM MobiSys 2023, ACM SenSys 2021, IEEE INFOCOM 2021

Publications

† *PhD/Master/Undergraduate students (co-)advised by me*; **Google Scholar Citations: 340+; h-index: 10**

[1] No Guide, No Cheat: Detecting Smart Glasses via AR Optical Signatures — ***ACM HotMobile***
Hanting Ye, Tianyi Hu, Maria Gorlatova. *2026*
- The International Workshop on Mobile Computing Systems and Applications.

[2] SHARE: Towards AR with User-Centric SLAM in Shared Human–Robot Workspaces — ***ACM IMWUT***
Tianyuan Du†, Tianyi Hu, **Hanting Ye**, Maria Gorlatova. *2026*
- The Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (CSRankings HCI venue).
- Code: SHARE.

[3] FlexSplat: Flexible Feed-Forward 3D Gaussian Splatting without Point Cloud Correspondence — ***BMVC***
Amir Sabbaghziarani†, **Hanting Ye**, Maria Gorlatova, Yi Ding. *2026*
- The British Machine Vision Conference.
- Project page and code.

[4] ARTOO-DARTU: Studying AR-Assisted Human-Robot Collaboration with AR Obstruction Mitigation during a Warehouse Task — ***ACM MobileHCI***
Christian Fronk†, **Hanting Ye**, Zhehan Qu, Maria Gorlatova. *2026*
- ACM International Conference on Mobile Human-Computer Interaction.

[5] fARfetch: Enabling Collocated AR-HRC in Large Visually Diverse Environments with VLM-Driven AR Content Adaptation — ***IEEE RO-MAN***
Christian Fronk†, **Hanting Ye**, David Hunt, Miroslav Pajic, Maria Gorlatova. *2026*
- IEEE International Conference on Robot and Human Interactive Communication.

[6] 5G × Headsets: An App-Level Measurement Study of 5G Latency for Networked Immersive XR — ***ACM ImmerCom***
Tianyuan Du†, **Hanting Ye**, Yiming Li, Tianyi Hu, Tingjun Chen, Maria Gorlatova. *2026*
- ACM Workshop on Mobile Immersive Computing, Networking, and Systems.
- Open-source setup: OAI-5G-Quest3-Tutorial.

[7] Real-Time Visual Obstruction Detection in Surgical Augmented Reality — ***IEEE ISMAR Medical XR***
Shih-Chin Yang†, Yanming Xiu, **Hanting Ye**, Qi Chen, Elias Rotondo, Maria Gorlatova. *2026*
- IEEE International Symposium on Mixed and Augmented Reality (ISMAR) Workshop on Medical XR.

[8] NIRF: Detecting Cameras That Hide Behind Screen — ***ACM MobiCom***
Hanting Ye, Niels van der Kolk†, Qing Wang. *2025*
- ACM Conference on Mobile Computing and Networking (CSRankings mobile computing venue).
- Acceptance rate: 17.1%

[9] Computing behind Transparent Screen — ***EWSN***
Hanting Ye, Qing Wang. *2024*
- International Conference on Embedded Wireless Systems and Networks.
- Acceptance rate: 22.2%

[10] Screen Perturbation: Adversarial Attack and Defense on Under-Screen Camera ***ACM MobiCom***

Hanting Ye, Guohao Lan, Jinyuan Jia, Qing Wang. *2023*

- ACM Conference on Mobile Computing and Networking (CSRankings mobile computing venue).
- Acceptance rate: 24.4%
- Code and data: Screen Perturbation.

[11] When VLC Meets Under Screen Camera ***ACM MobiSys***

Hanting Ye, Jie Xiong, Qing Wang. *2023*

- ACM Conference on Mobile Systems, Applications, and Services (CSRankings mobile computing venue).
- Acceptance rate: 20.7%
- Code: DynamicColor.

[12] SpiderWeb: Enabling Through-Screen Visible Light Communication ***ACM SenSys***

Hanting Ye, Qing Wang. *2021*

- ACM Conference on Embedded Networked Sensor Systems (CSRankings mobile computing venue).
- Acceptance rate: 18%
- Code and data: Through-Screen VLC.

[13] Optimization for Full-Duplex Rotary-Wing UAV-Enabled Wireless-Powered IoT Networks ***IEEE TWC***

Hanting Ye, Xin Kang, Jingon Joung, Ying-Chang Liang. *2020*

- IEEE Transactions on Wireless Communications

[14] Fingertip Air-Writing with Ambient Light ***EAI MobiQuitous***

Hao Liu†, **Hanting Ye**, Xiangrui Zhang, Jie Yang, Qing Wang. *2023*

- EAI Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services.

[15] Through-Screen Visible Light Sensing Empowered by Embedded Deep Learning ***AIChallengeIoT***

Hao Liu†, **Hanting Ye**, Jie Yang, Qing Wang. *2021*

- ACM SenSys Workshop on Challenges in Artificial Intelligence and Machine Learning for Internet of Things.

[16] Joint Uplink-and-Downlink Optimization of 3D UAV Swarm Deployment for Wireless-Powered IoT ***IEEE IoT-J***

Hanting Ye, Xin Kang, Jingon Joung, Ying-Chang Liang. *2021*

- IEEE Internet of Things Journal

[17] Optimization for Wireless-Powered IoT Networks Enabled by an Energy-Limited UAV Under Practical Energy Consumption Model ***IEEE WCL***

Hanting Ye, Xin Kang, Jingon Joung, Ying-Chang Liang. *2021*

- IEEE Wireless Communications Letters

[18] Cognitive Backscatter NOMA Networks With Multi-Slot Energy Causality ***IEEE CL***

Jie Wang*, **Hanting Ye***, Xin Kang, Sumei Sun, Ying-Chang Liang. *2020*

- IEEE Communications Letters
- *Equal contribution

[19] Joint Uplink and Downlink 3D Optimization of an UAV Swarms for Wireless-powered NB-IoT ***IEEE GLOBECOM'19***

Hanting Ye, Xin Kang, Jingon Joung, Ying-Chang Liang. *2019*

- IEEE Global Communications Conference

[20] Optimal Time Allocation for Full-duplex Wireless-Powered IoT Networks with UAV ***IEEE ICC'19***

Hanting Ye, Xin Kang, Jingon Joung, Ying-Chang Liang. *2019*

- IEEE International Conference on Communications

[21] Full-duplex Wireless-powered IoT Networks with Unmanned Aerial Vehicle ***IEEE ICTC'18***

Hanting Ye, Xin Kang, Ying-Chang Liang, Jingon Joung. *2018*

- IEEE International Conference on ICT Convergence
- Invited paper

Demonstrations

† *PhD/Master/Undergraduate students (co-)advised by me*

[1] Demo: WordPlay Party: Collaborative XR game for Parkinson's Rehab ***AWE USA***

Hanting Ye, Tianyuan Du†, James Finley, Jiasi Chen, Maria Gorlatova. *June 2026*

- Augmented World Expo USA; AWE 2026 poster spotlight.
- Long Beach, California

[2] Demo: WordPlay Party: Towards Collaborative XR App Latency Guarantees for PD Rehab ***UIUC IMMERSE***

Hanting Ye, Tianyuan Du†, James Yong†, Jiacheng Guo†, James Finley, Jiasi Chen, Maria Gorlatova. *April 2026*

- UIUC IMMERSE Symposium; symposium poster.
- Urbana, Illinois

[3] Demo: Catch Smart Glasses If You Scan: Robot-Controlled vs. UI-Guided ***ACM HotMobile***

Hanting Ye, Tianyi Hu, Valiantsin Kasabrukhau†, Sulaiman Khydyr uulu†, Maria Gorlatova. *February 2026*

- The International Workshop on Mobile Computing Systems and Applications.
- ***Best Demo Award***

[4] Demo: Control the ARmada: LMM Coordination for Multi-Robot AR-HRC ***ACM HotMobile***

Christian Fronk†, **Hanting Ye**, Miroslav Pajic, Maria Gorlatova. *February 2026*

- The International Workshop on Mobile Computing Systems and Applications.

Research Projects

Breaking Low - End-to-End Delivery Technology for Interactive Multi-person XR Rehabilitation *Duke*

Postdoctoral Research Associate

- Leading a multidisciplinary team, including 3 graduate game developers, 1 Ph.D. O-RAN researcher, and 2 physical therapy graduates, in developing a multi-person XR rehabilitation application on commercial XR platforms (Meta Quest 3 and VIVE Ultimate Trackers) for Parkinson's patients. The remote application provides dual-task training for individuals with both motor and cognitive impairments, enabling at-home rehabilitation under remote therapist supervision. A co-located multi-user demo of the *WordPlay* application has been released.
- Coordinating with academic (University of Michigan) and industry (Nokia) partners in the Breaking Low project to build ultra-low-latency networking infrastructure on O-RAN systems that support immersive XR experiences for Parkinson's patients using *WordPlay*.
- Collaborating with clinical partners at the University of Southern California to develop a patient tracking data collection pipeline and to recruit Parkinson's patients for an IRB-approved user study to begin real-world evaluation.
- The XR rehabilitation app, WordPlay Party, was distributed to multiple collaborators and clinics and demonstrated to therapists.
- Research Output: 2 Demos @AWE USA 2026 @UIUC IMMERSE; 1 Paper @ ImmerCom'26; 1 Feature Page in Duke University Center for Computational and Digital Health Innovation.

AI and AR for Human-Robot Collaboration *Duke*

Postdoctoral Research Associate

- Co-advising a PhD student on developing edge-assisted systems that allow AR content to display robots' internal states without obstructing important real-world objects. Also guiding efforts to leverage VLMs to enhance the legibility of AR content in large-scale and complex environments as part of DARPA-funded projects.
- Co-advising a PhD student on designing new approaches to reduce communication and AI computation overhead in existing multi-agent SLAM systems in order to meet strict latency constraints for AR visual cues. These cues provide spatial awareness for heterogeneous agent groups (humans and robots) in human-robot collaboration contexts.
- Secured new research funding from NSF CORE in Human-Centered Computing and Meta, i.e., the equipment grant for five Meta Aria Gen 2 glasses to facilitate human-robot collaboration.
- Research Output: 1 Paper @ IMWUT'26; 1 Paper @ MobileHCI'26; 1 Paper @ RO-MAN'26.

Latency Optimization for VLM Inference on Smart Glasses in Medical Applications *Duke*

Postdoctoral Research Associate

- Jointly supervised one undergraduate student and two master's students with a PhD student to optimize VLM inference latency through edge-cloud collaborative computing, including VLM attention map-based token pruning, temporal frame selection from captured AR frames, and an early-exit mechanism guided by occlusion cues of critical real-world objects.
- Leveraged video generation models, including Seedance and Kling, to develop a real-time safety detection system for smart glasses in ICU and operating-room environments, aiming to prevent AR content from occluding critical real-world objects, such as surgical instruments and organs.
- Research Output: 1 Paper @MedicalXR, 2026.

Detecting AI smart glasses with Invisible AR Waveguides *Duke*

Postdoctoral Research Associate

- Identified **eye glow** and **rainbow patterns**, two core optical signatures in 10%–30% of captured frames through comprehensive scans of diffractive-waveguide AR displays across Even, RayNeo, StarV, Snap, and Quark smart glasses.
- Jointly supervised two undergraduate students with another PhD student in designing and building an automated 4-DoF optical-characterization platform for systematically scanning diffractive-waveguide AR displays under human-like head rotations (140° yaw, 120° pitch, and 80° roll), capturing more than 1,000 eyeglass-scanning videos, and developing a smartphone app that enables AR smart glasses detection using off-the-shelf smartphones.
- 90+% accuracy in distinguishing AR smart glasses from ordinary eyeglasses by detecting neural-network-estimated optical-flow inconsistencies caused by AR waveguide-induced brightness variations.
- Research Output: 1 Paper and 1 Demo @ ACM HotMobile'26 (Best Demo Award); 1 Media quiry from Korea Media EToday.

Privacy Protection of Under-Screen Camera *TU Delft*

PhD student

- Developed an optical model to simulate various screen pixel layouts under active light interference, assessing the effects of pixel size, shape, and arrangement on under-screen camera imaging.
- Developed and implemented 'One-Pixel Perturbation' and 'Multi-Pixel Perturbation' strategies, achieving over 90% reduction in neural network recognition accuracy for images, thus significantly enhancing key privacy safeguards for sensitive information like facial data.
- Led the conceptualization and mentoring of student projects focused on using retroreflections and time-of-flight sensors for the detection of hidden under-screen cameras, fostering practical skills and research methodologies.
- Research Output: 1 Paper @ MobiCom'23; 1 Paper @ MobiCom'25; 1 News Report in Department.

Enabling Through-Screen Visible Light Communication (VLC) *TU Delft*

PhD student & Visiting researcher at PureLiFi

- Pioneered the concept of through-screen VLC, demonstrating its practical feasibility and laying the groundwork for future advancements in optical communications.
- Identified and analyzed a color-pulling effect in transparent OLED screens impacting color-based CSK signals; developed a novel pixel-sweeping algorithm to mitigate rolling-shutter effects in under-screen cameras.
- Designed a spiderweb-inspired 'SWebCSK' modulation scheme, integrating a color-based approach with a unique slope-boosting demodulation technique, significantly enhancing signal clarity and transmission efficiency.
- Research Output: 1 Paper @ MobiSys'23; 1 Paper @ SenSys'21; 1 Public Outreach in European Researchers' Night.

Embedded Deep Learning Empowered Fingertip Air-Writing with Ambient Light *TU Delft*

Co-advisor of MSc student

- Innovatively designed and prototyped an ambient light-based fingertip air-writing system.
- Conducted a comprehensive analysis of in-air digit writing patterns, leading to the creation of a unique dataset for air-written digits.
- Developed a novel calibration method to significantly improve the system's robustness and accuracy in diverse lighting conditions.
- Designed two advanced embedded deep learning models to decode spatial and temporal patterns in dynamic shadows.
- Research Output: 1 Paper @ AIChallengeIoT'21; 1 Paper @ MobiQuitous'23.

Resource Allocation and Optimization of Time/Power-Limited UAV Networks *UESTC*

Master student

- Devised an innovative time allocation algorithm to optimize system performance under sparse user distribution scenarios, leveraging the interplay between UAV hovering and flight times for enhanced efficiency.
- Designed a comprehensive framework for dense user environments, incorporating joint optimization of 3D UAV positioning, device-UAV associations, and scheduling for both uplink and downlink, significantly boosting network throughput and reliability.
- Research Output: 1 Paper @ TWC; 1 Paper @ IoT-J; 1 Paper @ WCL; 1 Paper @ CL; 1 Paper @ GLOBECOM; 1 Paper @ ICC; 1 Paper @ GLOBECOM; 1 Paper @ ICTC.

Teaching Experience

ECE/COMPSCI 564: *Project Mentor*, *Edge Computing* [5 credits], MSc course, Spring, 2026

ECE/COMPSCI 356: *Alternate Lecturer*, *Computer Network Architecture* [5 credits], BSc course, Winter, 2025

CSE2425: *Teaching assistant*, *Embedded Software* [5 credits], BSc course, Q2, 2020, 2021, 2022, 2023

CESE4110: *Teaching assistant*, *Visible Light Communication and Sensing* [5 credits], MSc course, Q4, 2021-2022, Q1, 2023-2024

Selected Research Supervision

Independently and jointly mentored **20 students, including 2 K–12 students, 7 undergraduates, 8 graduate students, and 3 PhD students** at Duke, GSU, and TU Delft, resulting in **11 publications**. One K–12 mentee was admitted to the **University of Toronto**, one Duke undergraduate earned **Graduating with Distinction**, and two TU Delft master's students earned **8.5/10**. Work by an independently supervised master's student contributed to an **ACM MobiCom 2025** paper, and work by two supervised undergraduates contributed to the **ACM HotMobile 2026 Best Demo Award**.

Tianyuan Du, PhD Student, Low-Latency Guarantees in AR Human-Robot Collaboration and Networked AR Healthcare Systems *Duke*

I am his co-supervisor *05/2025–Present*

- I mainly guided the student in identifying that AR users exhibit varying sensitivity to virtual content and robot tracking in human-robot collaborative scenarios. He developed an adaptive algorithm to address imbalanced allocation of communication and computational resources across heterogeneous agents (humans and robots), enabling the edge-based SLAM system to maintain spatial positioning within a strict 20 ms latency constraint. This work has been published at **ACM IMWUT**. I am also leading him in investigating cross-XR applications and O-RAN multi-user Parkinson's rehabilitation system design to guarantee low user-to-user latency in the **NSF Breaking Low project**.

Christian Fronk, PhD Student, Enhancing the Safety of AR Content in Human-Robot Collaboration (HRC) Scenarios *Duke*

I AM HIS CO-SUPERVISOR *05/2025–Present*

- I mainly guided the student in improving the visibility of critical real-world objects and the legibility of AR content within the user's field of view in HRC scenarios. These works focused on ensuring that virtual displays, such as those visualizing the robot's internal state via an edge-computing-supported object detection pipeline, do not obscure key physical elements (e.g., safety-related signage in a warehouse). Semantic segmentation and vision-language model capabilities are also leveraged to enhance the clarity of virtual content, such as robot target locations, in outdoor and complex environments. These works are published at **IEEE RO-MAN** and **ACM MobileHCI**.

Valiantsin Kasabrukhau, Undergraduate Student, Robot-arm-based, controllable scanning platform with user-customizable trajectories *Duke*

I AM HIS CO-SUPERVISOR *12/2025–Present*

- I mainly guided the student in leveraging the robot arm to enable a reproducible smart glasses scanning process, where scanning trajectories can be customized by users. The scanning platform consists of a 4-DoF robot (PincherX 100 from Trossen Robotics), a 3D-printed face mask for mounting the smart glasses, and a smartphone camera on a stand. For robot control, we use a Linux-based Beelink Mini S12 running ROS2. We developed a UI with 11 predefined basic robot arm movements, which can be combined and adjusted in terms of the order and amplitude of each movement to mimic human head motions within ranges of 140° yaw, 120° pitch, and 80° roll. The resulting demonstration was presented at **ACM HotMobile** and received the **Best Demo Award**.

Sulaiman Khydyr uulu, Undergraduate Student, Smartphone-based scanning pipeline with UI indicators *Duke*

I AM HIS CO-SUPERVISOR *12/2025–Present*

- I mainly guided the student in developing a detector app on a Google Pixel 3 with UI indicators to guide users during scanning, ensuring that the smart glasses are observed from a comprehensive range of angles. We deploy the Google ML Kit face detection model to identify facial landmarks and use them to calculate the spatial orientation of the user's head pose. To simplify the user guidance experience, we create a ring around the user composed of 80 short grey bars. These bars represent different facial orientations. When a grey bar is covered for a certain period of time, it turns green. We also display instructional text at the bottom of the app to guide users to move toward any uncovered orientations until all the grey bars turn green. The resulting demonstration was presented at **ACM HotMobile** and received the **Best Demo Award**.

Adrian Willet, K-12 Student, Gaussian Streaming: Visualizing Worlds on Mobile XR Devices *Duke*

I SUPERVISED HIM INDEPENDENTLY *06/2025–Present*

- I mainly supervise him in developing a new Gaussian Splatting technique to help mobile XR devices, such as the Meta Quest 3, load reconstructed 3D scenes. The final research results were presented at the North Carolina School of Science and Mathematics (NCSSM) Summer Research & Innovation Program presentation. He is now a student at the **University of Toronto**.

Niels van der Kolk, Master Student, Under-Screen Camera Detection *TU Delft*

I SUPERVISED HIM INDEPENDENTLY *06/2022–07/2023*

- Under my supervision, he developed the world's first under-screen camera detection system using a ToF sensor. This innovative project addresses growing concerns about the potential privacy risks posed by hidden spy cameras, highlighting our commitment to leveraging technology for societal well-being. The related work contributed to a paper published at **ACM MobiCom**. His final grade is **8.5/10**.

Hao Liu, Master Student, LightDigit: Embedded Deep Learning Empowered Fingertip Air-Writing with Ambient Light *TU Delft*

I AM HIS CO-SUPERVISOR *10/2020–08/2021*

- He is now a PhD student at TU Delft. Under my co-supervision (together with Prof. Q. Wang and Prof. J. Yang), he has published a paper at **ACM AIChallengeIoT@SenSys** and a paper at **EAI MobiQuitous**. His final grade was **8.5/10**.

Professional Service

National funding agency service

- US National Science Foundation (NSF): proposal review panelist, communications and networking, 2025

Technical Program Committees (TPCs) for Conferences

- The 33rd ACM Conference on Mobile Computing and Networking (MobiCom'27)
- The 25th ACM Conference on Embedded Networked Sensor Systems (SenSys'27)
- The 30th Design, Automation and Test in Europe Conference (DATE'27)
- The 24th ACM Conference on Mobile Systems, Applications, and Services (MobiSys'26)
- The 24th ACM Conference on Embedded Networked Sensor Systems (SenSys'26)
- The 23rd International Conference on Embedded Wireless Systems and Networks (EWSN'26)
- The 29th Design, Automation and Test in Europe Conference (DATE'26)
- The 35th International Conference on Computer Communications and Networks (ICCCN 2026)
- The 23rd ACM Conference on Embedded Networked Sensor Systems (SenSys'25)
- The 22nd International Conference on Embedded Wireless Systems and Networks (EWSN'25, **Distinguished TPC Member**)
- The 39th Annual AAAI Conference on Artificial Intelligence (AAAI'25)
- The second Workshop on Enhancing Security, Privacy, and Trust in Extended Reality (XR) Systems (XRSecurity'26)
- The first Workshop on Enhancing Security, Privacy, and Trust in Extended Reality (XR) Systems (XRSecurity'25)

Organizing Committee

- Web Co-Chair, EWSN'25
- Local Arrangements Co-Chair, EWSN'21
- Student Volunteer, ACM SIGCOMM'22
- Student Volunteer, IEEE ICNP'21
- Student Volunteer, IEEE ICCS'18

Reviewers for Conferences

- The 22nd Annual ACM/IEEE International Conference on Human-Robot Interaction (HRI'27)
- The 33rd IEEE Conference on Virtual Reality and 3D User Interfaces (VR'27)
- The 40th Annual Conference on Neural Information Processing Systems (NeurIPS'26)
- The 2026 ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT'26)
- The 2025 ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT'25)
- The 32nd IEEE Conference on Virtual Reality and 3D User Interfaces (VR'26)
- The 40th Annual Conference on Neural Information Processing Systems (NeurIPS'26)
- The 18th European Conference on Computer Vision (ECCV'26)
- The 2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR'26)
- The 24th IEEE International Symposium on Mixed and Augmented Reality (ISMAR'25)
- The 2025 IEEE/CVF International Conference on Computer Vision (ICCV'25)
- The 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR'25)
- The 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR'24)

Reviewers for Journals

- ACM Transactions on Sensor Networks (TOSN, **Distinguished Reviewer Board**)
- IEEE Journal on Selected Areas in Communications (JSAC)
- IEEE Transactions on Wireless Communications (TWC)
- IEEE Transactions on Mobile Computing (TMC)
- IEEE Transactions on Services Computing (TSC)
- IEEE Transactions on Communications (TCOM)
- IEEE Internet of Things Journal (IoT-J)
- IEEE Communications Letters (CL)
- IEEE Wireless Communications Letters (WCL)

External Reviewers for Conferences and Journals

- 2026: ACM MobiCom
- 2025: IEEE INFOCOM, ACM MobiSys
- 2024: IEEE INFOCOM, ACM MobiCom
- 2023: IEEE INFOCOM, IEEE SECON
- 2022: IEEE INFOCOM, ACM SenSys
- 2021: IEEE INFOCOM
- 2017-2020: IEEE TWC, IEEE TCOM, IEEE TVT, IEEE IoT-J

Outreach and Event Organization

- *Lead Organizer*: Duke I3T Lab, Durham, USA, *Events:* Open Lab, E4USA, and Demo Days.
- *Public Outreach*: 10/2022, Vienna, Austria, *Event:* European Researchers' Night; *Title:* Can LEDs talk?

Lectures, Presentations, and Talks

- *Lecturer*: 11/2026, Pennsylvania State University, USA, *Title:* Towards Physically Augmented Mobile AI.
- *Lecturer*: 02/2026, Duke University, USA, *Title:* Using Edge Computing to Enhance Foundation Models.
- *Conference Presentation*: 02/2026, Atlanta, USA, *Title:* No Guide, No Cheat: Detecting Smart Glasses via AR Optical Signatures.
- *Conference Presentation*: 11/2025, Hong Kong, China, *Title:* NIRF: Detecting Cameras That Hide Behind Screen.
- *Conference Presentation*: 12/2024, Abu Dhabi, United Arab Emirates, *Title:* Computing behind Transparent Screen.
- *Conference Presentation*: 10/2023, Madrid, Spain, *Title:* Screen Perturbation: Adversarial Attack and Defense on Under-Screen Camera.
- *Conference Presentation*: 06/2023, Helsinki, Finland, *Title:* When VLC Meets Under-Screen Camera.
- *Talk*: 06/2023, Soesterberg, The Netherlands, *Title:* When VLC Meets Under-Screen Camera.
- *Talk*: 10/2022, Linz, Austria, *Title:* Through-Screen LiFi System.
- *Conference Presentation*: 11/2021, Coimbra, Portugal, *Title:* SpiderWeb: Enabling Through-Screen Visible Light Communication.
- *Conference Presentation*: 12/2019, Hawaii, USA, *Title:* Joint Uplink and Downlink 3D Optimization of an UAV Swarms for WP NB-IoT.
- *Conference Presentation*: 05/2019, Shanghai, China, *Title:* Optimal Time Allocation for Full-duplex Wireless-Powered IoT Networks with UAV.

Media Coverage

- *Duke University Center for Computational and Digital Health Innovation*: 09/2026, Feature Page *Link*: *Featured Researcher: Hanting Ye*
- *EEMCS Department of TU Delft*: 11/2025, LinkedIn Post *Link*: *TU Delft's Embedded AI Lab for developing a new system that can detect behind-screen hidden cameras!*